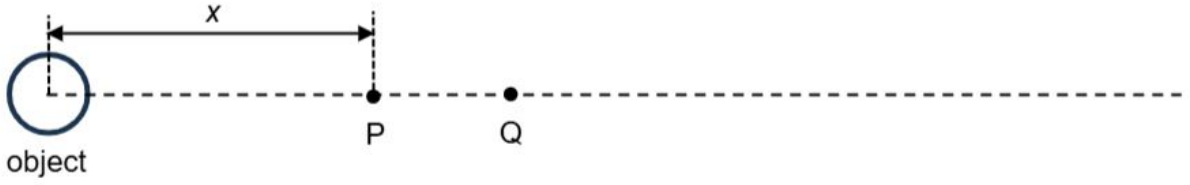
12 Two points P and Q are a fixed distance apart on a line joining them to an object considered as a point mass. Point P is at a distance, x away from the object. The magnitude of the gravitational potential at point P is ϕ.



The two points move away from the object, along the line, maintaining their fixed separation such that point P is now at a distance $2x$ away from the object.

What happens to the magnitude of the gravitational potential at the new point P and to the magnitude of the gravitational potential difference between point P and point Q?

	magnitude of gravitational potential at the new point P	magnitude of gravitational potential difference
A	less than $\frac{\phi}{2}$	decreases
B	less than $\frac{\phi}{2}$	stays the same
C	$\frac{\phi}{2}$	decreases
D	$\frac{\phi}{2}$	stays the same

L2 **Answer: C**

$$\textit{gravitational potential} = -\frac{GM}{r}$$

GM is constant. Hence **when distance r from object of mass M is doubled, magnitude of gravitational potential at new point P is exactly halved.**
→ Answer is either option C or D.

Additionally, **due to the inverse-proportional relationship between potential and r (hyperbolic curve), when both the distances for points P and Q increased by the same distance x, the magnitude of the potential at point P (where distance from object is smaller) decreases by a greater amount than the decrease in potential at point Q (where distance from object is greater). As such, the difference between the potentials at P and Q decreases.**
→ Answer is C not D.